

SNA Fall 2025: Assignment 3

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Libraries and Data

Libraries

```
library(igraph)
library(dplyr)
library(intergraph)
library(ggplot2)
library(tidyverse)
library(ergm)
library(sna)
library(car)
library(knitr)
library(kableExtra)
```

Data

```
load("luke.datasets.RData")
```

Explore

```
df <- data.frame(DHHS)

kable(head(df, 10),
      format = "latex",
      booktabs = TRUE,
      caption = "DHHS Network (first 10 rows)" %>%
      kable_styling(full_width = TRUE,
                    latex_options = "HOLD_position")
```

Table 1: DHHS Network (first 10 rows)

.tail	.head	collab
ACF-1	ACF-2	1
ACF-2	CDC-6	2
ACF-2	CDC-9	1
ACF-2	OS-3	2
ACF-2	OS-4	1
ACF-2	OS-5	3
AHRQ-1	AHRQ-2	3
AHRQ-1	AHRQ-3	3
AHRQ-1	AHRQ-4	3
AHRQ-1	CDC-3	1

Task 1: Network Description and Visualization

DHHS

Description

```
describe.DHHS()
```

The DHHS Collaboration Network represents communication and collaboration among 54 tobacco control leaders across 11 agencies within the U.S. Department of Health and Human Services (DHHS) in 2005. Each node is a DHHS tobacco control leader, and an undirected tie indicates regular contact or collaboration about tobacco control issues (at least quarterly). The data were collected through a cross-sectional survey described by Leischow et al. (2010), which examined how agencies such as the CDC, NIH, and the Office of the Secretary exchanged information to coordinate national tobacco control efforts.

Details

```
class(DHHS)
```

```
## [1] "network"
```

```
network.size(DHHS)
```

```
## [1] 54
```

```
network.edgecount(DHHS)
```

```
## [1] 447
```

```
is.directed(DHHS)
```

```
## [1] FALSE
```

```
list.vertex.attributes(DHHS)
```

```
## [1] "agency"      "na"          "vertex.names"
```

```
list.edge.attributes(DHHS)
```

```
## [1] "collab" "na"
```

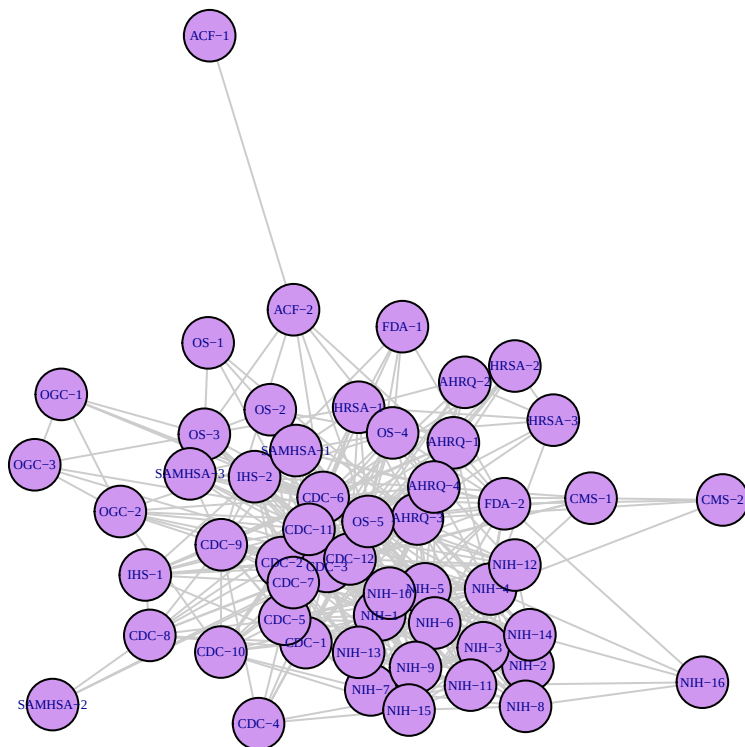
Visualization

```
dhhs <- graph_from_data_frame(d = DHHS, directed = FALSE)  
class(dhhs)
```

```
## [1] "igraph"
```

```
set.seed(42)  
plot(dhhs,  
     vertex.color = "#cf97f0",  
     edge.color = "#cccccc",  
     vertex.label.cex = 0.4,  
     main = "DHHS Collaboration Network")
```

DHHS Collaboration Network



```
dhhs_orgs <-  
  data.frame(node1 = str_extract(df$.tail, pattern = "[A-Z]{1,5}"),  
            node2 = str_extract(df$.head, pattern = "[A-Z]{1,5}")) %>%  
  group_by(node1, node2) %>%
```

```

summarize(weight = n(), .groups = "drop")

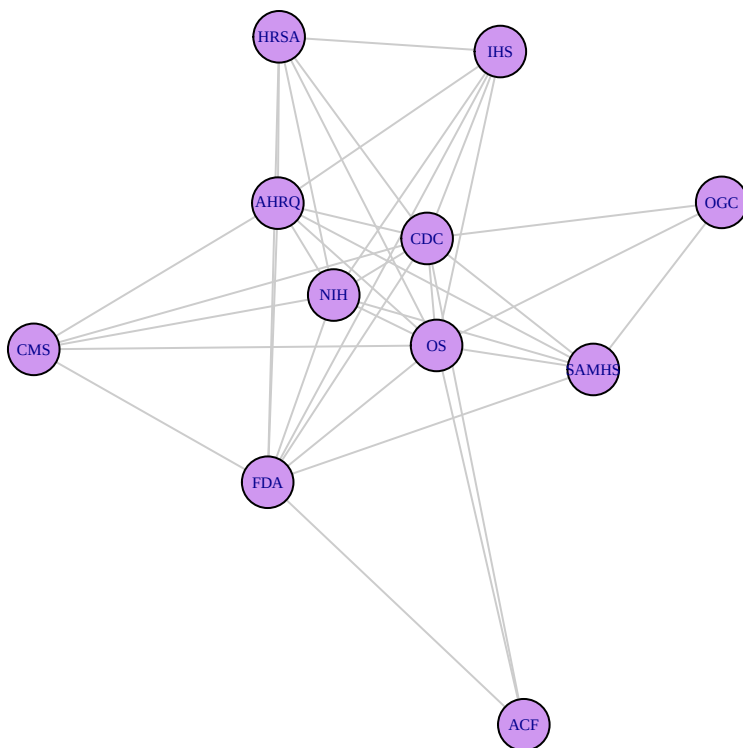
dhhs_orgs_gr <- graph_from_data_frame(dhhs_orgs, directed = FALSE)

set.seed(42)
gr <- igraph::simplify(dhhs_orgs_gr)

plot(gr,
     vertex.color = "#cf97f0",
     edge.color = "#cccccc",
     vertex.label.cex = 0.5,
     main = "DHHS Collaboration Network"
)

```

DHHS Collaboration Network



Middle School

Description

```
describe.middleschool()
```

The Middle School Friendship Network represents directed friendship ties among a class of 6th-grade students in California. Each student could nominate up to five closest friends in their class, producing a directed network with 37 vertices (students) and 179 arcs (friendship nominations). The network includes one node-level attribute, gender (1 = female, 2 = male). These data originate from Thomas Valente's *Social Networks and Health* (2010), where the network was used to study social influence and peer relationships in adolescent health and behavior contexts.

Details

```

class(middleschool)

## [1] "network"
network.size(middleschool)

## [1] 37
network.edgecount(middleschool)

## [1] 179
is.directed(middleschool)

## [1] TRUE
list.vertex.attributes(middleschool)

## [1] "gender"      "na"          "vertex.names"
list.edge.attributes(middleschool)

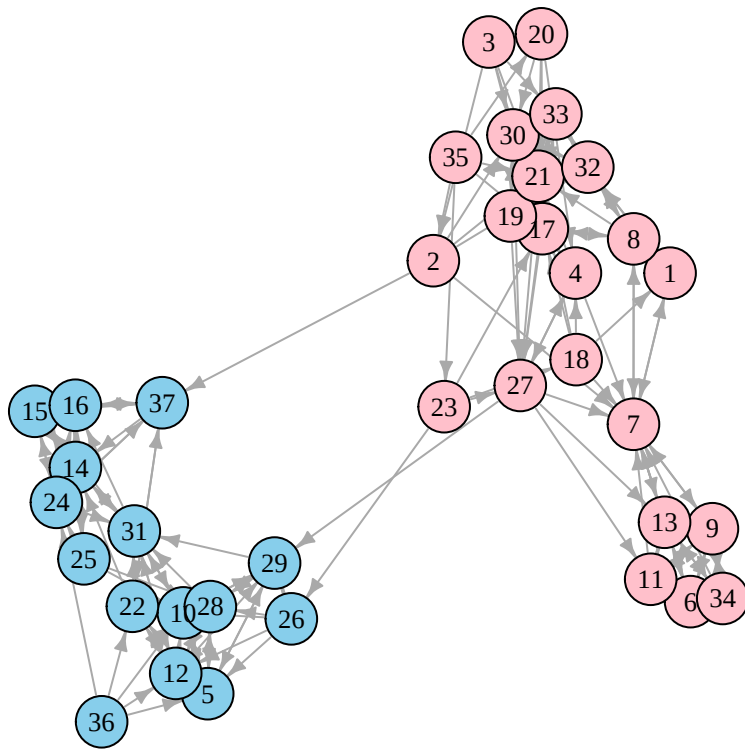
## [1] "na"
Visualization
set.seed(42)

g_middleschool <- graph_from_data_frame(d = middleschool, directed = TRUE)
V(g_middleschool)$gender <- middleschool %v% "gender"
V(g_middleschool)$color <- ifelse(V(g_middleschool)$gender == "2", "skyblue", "pink")

plot(g_middleschool,
     vertex.label.color = "black",
     vertex.label.cex = 0.8,
     vertex.color = V(g_middleschool)$color,
     edge.arrow.size = 0.5,
     main = "Middle School Friendship Network")

```

Middle School Friendship Network



Task 2: Homophily Descriptive Measures - Blau Index

Define (~200 words)

- The Blau Index (also called the Diversity Index) measures the probability that two randomly selected individuals belong to different categories (e.g., race, ethnicity, or gender) (Lecture 5 Slides, p.27). The index ranges from 0 (homophily, no diversity) to 1 (heterophily, full diversity).
- The Blau Index assumes that only the degree of heterogeneity matters, and that outcomes are completely neutral to which specific groups are large or small, meaning populations with different racial mixes can be treated as equally “diverse” as long as their proportions generate the same heterogeneity score (Rushton, 2008). It also ignores social meaning or power asymmetry of categories, implicitly assuming individuals from all groups contribute equally to overall heterogeneity (Rushton, 2008, p.447). Its strength is simple, intuitive, and comparable across contexts (Rushton, 2008, p.446–447), useful when researchers only care about the degree of mixture, not which groups dominate. However, it ignores which specific groups are large or small, treating very different group compositions as equally “diverse”, and thus loses important substantive information.
- It is a local measure and applied to ego-network. In SNA, the Blau Index is mostly used for ego-networks and is computed per each node using only direct neighbors of ego (Lecture 5 Slides, p.29). Blau Index gives a score for every node: the expected output is a score for each node in the network (Lecture 5 Slides, p.30)
- In social network analysis (SNA), the Blau Index is most appropriate when the research question concerns heterogeneity or mixing patterns. However, it is not ideal when group identity, hierarchy, or power asymmetry matters (e.g., racial dominance, gender inequality). In such cases, we should supplement or replace Blau with measures that account for which groups interact (e.g., E-I Index or

assortativity).

Compute

- a. Write code to calculate Blau Index on this toy network. Write your code preferably in one or more functions, with clear documentation. Our implementation takes an igraph object as input (e.g., the object labeled as `network` in the code below) and uses igraph functions, such as `neighbors()`.

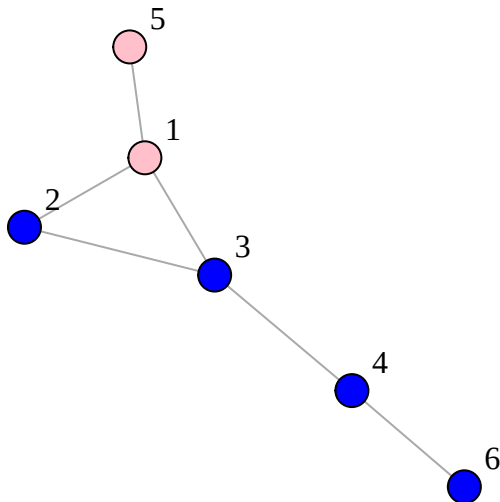
```
library(igraph)

df <- data.frame(
  from = c(1,2,3,4,5,6),
  to = c(2,3,1,3,1,4),
  gender = c("female", "male", "male", "male", "female", "male"))
df

##   from to gender
## 1    1  2 female
## 2    2  3  male
## 3    3  1  male
## 4    4  3  male
## 5    5  1 female
## 6    6  4  male

network <- graph_from_data_frame(df, directed = FALSE)
V(network)$gender <- df$gender

V(network)$color <- ifelse(V(network)$gender == "male", "blue", "pink")
plot(network,
      vertex.label.color = "black",
      vertex.label.dist = 2.5,
      vertex.color = V(network)$color)
```



```
toy_blau_index <- function(node) {
  # identify neighbors of the target node
  neighbors <- neighbors(network, node)
  # identify if there are any isolated nodes
  if (length(neighbors) == 0) {return(NA_real_)}
  # extract the categorical attribute values (here: gender) of the neighbors
  gender_values <- V(network)$gender[neighbors]
```

```

# compute the proportion of each category among the neighbors
gender_proportions <- table(gender_values) / length(gender_values)
# apply the Blau Index formula:  $1 - \sum(p_k^2)$ 
blau_index <- 1 - sum(gender_proportions^2)
# return the Blau Index value for this node
return(blau_index)
}

# apply the function to all nodes in the network
toy_blau_indexes <- sapply(1:vcount(network), toy_blau_index)
# display the Blau Index for each node
toy_blau_indexes

```

```
## [1] 0.4444444 0.5000000 0.4444444 0.0000000 0.0000000 0.0000000
```

- b. Normalize the scores, using the IQV or Index of Qualitative Variation: `iqv <- (blau_index / (1 - (1 / levels)))` where levels are the levels of your node attribute; for example, our toy networks has two levels, male and female.

```

# calculate the number of gender categories
levels <- length(unique(V(network)$gender))
# apply IQV normalization
toy_iqv <- toy_blau_indexes / (1 - (1 / levels))
# display IQV scores
toy_iqv

```

```
## [1] 0.8888889 1.0000000 0.8888889 0.0000000 0.0000000 0.0000000
```

- c. Apply your code to the network you are working with.

```

dhhs <- asIgraph(DHHS)

dhhs_blau_index <- function(node) {
  neighbors <- neighbors(dhhs, node)
  if (length(neighbors) == 0) {return(NA_real_)}
  agency_values <- V(dhhs)$agency[neighbors]
  agency_proportions <- table(agency_values) / length(agency_values)
  blau_index <- 1 - sum(agency_proportions^2)
  return(blau_index)
}

dhhs_blau_indexes <- sapply(1:vcount(dhhs), dhhs_blau_index)
sort(dhhs_blau_indexes)

```

```

## [1] 0.0000000 0.0000000 0.2400000 0.3140496 0.3200000 0.4444444 0.4444444
## [8] 0.4462810 0.4650000 0.4852071 0.4986150 0.5000000 0.5244444 0.5306122
## [15] 0.5714286 0.5800000 0.6065089 0.6074380 0.6172840 0.6176000 0.6222222
## [22] 0.6250000 0.6315789 0.6373365 0.6388889 0.6400000 0.6400000 0.6400000
## [29] 0.6575964 0.6755556 0.6777778 0.6802721 0.6938776 0.6938776 0.6979167
## [36] 0.7036011 0.7346939 0.7346939 0.7410208 0.7448980 0.7500000 0.7755102
## [43] 0.7755102 0.7755102 0.7777778 0.7839335 0.7889273 0.7944215 0.7968750
## [50] 0.8055556 0.8266667 0.8344671 0.8373702 0.8429752

```

Nodes with no neighbors (i.e., isolated nodes) would return a Blau Index of NA, but the computed vector contains no NA values, so the Blau Index values of 0 do not reflect undefined cases due to isolation. Instead, they represent nodes that do have neighbors, but all of those neighbors belong to the same agency, resulting in perfect local homophily.


```

levels <- length(unique(V(dhhs)$agency))
dhhs_iqv <- dhhs_blau_indexes / (1 - (1 / levels))
sort(dhhs_iqv)

## [1] 0.0000000 0.0000000 0.2640000 0.3454545 0.3520000 0.4888889 0.4888889
## [8] 0.4909091 0.5115000 0.5337278 0.5484765 0.5500000 0.5768889 0.5836735
## [15] 0.6285714 0.6380000 0.6671598 0.6681818 0.6790123 0.6793600 0.6844444
## [22] 0.6875000 0.6947368 0.7010702 0.7027778 0.7040000 0.7040000 0.7040000
## [29] 0.7233560 0.7431111 0.7455556 0.7482993 0.7632653 0.7632653 0.7677083
## [36] 0.7739612 0.8081633 0.8081633 0.8151229 0.8193878 0.8250000 0.8530612
## [43] 0.8530612 0.8530612 0.8555556 0.8623269 0.8678201 0.8738636 0.8765625
## [50] 0.8861111 0.9093333 0.9179138 0.9211073 0.9272727

library(tibble)

node_names <- V(dhhs)$vertex.names
node_agency <- V(dhhs)$agency

dhhs_blau_df <- data.frame(
  node = node_names,
  agency = node_agency,
  blau_index = dhhs_blau_indexes,
  iqvl      = dhhs_iqv )

dhhs_blau_df_sorted <- dhhs_blau_df[order(dhhs_blau_df$blau_index), ]
dhhs_blau_df_sorted <- rownames_to_column(dhhs_blau_df_sorted, var = "row")

kable(head(dhhs_blau_df_sorted, 10),
  format = "latex",
  booktabs = TRUE,
  caption = "DHHS Blau Index (first 10 rows)" %>%
  kable_styling(full_width = TRUE,
    latex_options = "HOLD_position")

```

Table 2: DHHS Blau Index (first 10 rows)

row	node	agency	blau_index	iqv
1	ACF-1	0	0.0000000	0.0000000
53	SAMHSA-2	10	0.0000000	0.0000000
35	NIH-8	7	0.2400000	0.2640000
26	IHS-1	6	0.3140496	0.3454545
43	NIH-16	7	0.3200000	0.3520000
30	NIH-3	7	0.4444444	0.4888889
41	NIH-14	7	0.4444444	0.4888889
14	CDC-8	2	0.4462810	0.4909091
38	NIH-11	7	0.4650000	0.5115000
29	NIH-2	7	0.4852071	0.5337278

```

btw_n <- igraph::betweenness(dhhs, normalized = TRUE)

btw_n_df <- data.frame(
  node = node_names,

```

```

agency      = node_agency,
betweenness = btw_n)

btw_n_df_sorted <- btw_n_df[order(btw_n_df$betweenness), ]

kable(head(btw_n_df_sorted, 10),
      format = "latex",
      booktabs = TRUE,
      caption = "DHHS Betweenness (first 10 rows)" %>%
      kable_styling(full_width = TRUE,
                    latex_options = "HOLD_position"))

```

Table 3: DHHS Betweenness (first 10 rows)

	node	agency	betweenness
1	ACF-1	0	0.0000000
53	SAMHSA-2	10	0.0000000
47	OS-1	9	0.0001382
20	CMS-2	3	0.0001451
43	NIH-16	7	0.0002117
29	NIH-2	7	0.0002873
44	OGC-1	8	0.0003749
46	OGC-3	8	0.0003840
54	SAMHSA-3	10	0.0003897
21	FDA-1	4	0.0003971

Actors such as ACF-1 and SAMHSA-2 have Blau Index = 0, which means all of their neighbors come from the same agency, they participate exclusively in intra-agency connections and show perfect local homophily. At the same time, their betweenness centrality = 0, meaning they do not sit on any shortest path connecting other parts of the network. Taken together, this indicates that these actors function as peripheral, agency-bounded participants: they belong to isolated clusters.

Interpret

The Blau Index measures local diversity—the probability that two randomly selected nodes in a network belong to different groups. In my DHHS network, I computed this index using the node attribute agency. The results show that the ego-level Blau scores range from 0 to about 0.84, indicating substantial variation in local heterogeneity across nodes.

Some nodes have 0 or very low values, meaning all or most of their connections are within the same agency. In the context of the DHHS findings, these egos correspond to members of agencies that were structurally isolated in the original study: agencies like CMS or FDA, whose tobacco control leaders interacted primarily with one another and rarely with others. From the ego’s perspective, this reflects a locally isolated communication environment in which cross-agency collaboration opportunities are absent, reinforcing the system-wide gaps identified in the paper.

In contrast, most other egos have high Blau scores (exceed 0.6), suggesting highly mixed connections that bridge multiple agencies. Especially those in CDC and OS, which the paper identified as major bridging agencies, have neighbors from multiple agencies and thus operate in richly heterogeneous local networks. At the ego-level, such high Blau values suggest that these individuals experience the DHHS system as integrative and cross-boundary, consistent with their observed roles as brokers and coordinators in the paper analysis.

After normalization (IQV), the pattern remains: most values fall between roughly 0.5 and 0.9, indicating that while a few nodes interact mainly within their own organizational boundary, the majority maintain highly

diverse inter-agency ties. That means, a generally high level of inter-agency heterogeneity. In the DHHS framework, these egos are precisely the individuals who help compensate for structural silos by connecting otherwise disconnected agencies. Their high local diversity is a micro-level reflection of the bridging functions that the paper analysis attributed to central CDC, OS, and NIH actors.

Substantively, this pattern reflects status homophily rather than value homophily in Lazarsfeld & Merton's (1954) sense. Since "agency" represents an ascribed institutional status rather than beliefs or values, high Blau scores reveal nodes that cross structural boundaries, whereas low scores indicate status-based clustering. In McPherson et al.'s (2001) framework, such tendencies are largely baseline (induced) homophily, shaped by structural opportunities: individuals in the same agency are organizationally collocated and thus more likely to form ties.

However, the results show that most nodes have relatively high Blau Index values (often above 0.5), indicating a high degree of local heterogeneity. This suggests that, despite the inherent tendency toward status and baseline homophily, many actors actively engage in cross-agency interactions. In other words, the network is not fully segmented along institutional lines-while the structure creates natural clusters within agencies, most individuals still maintain diverse, inter-agency connections that facilitate coordination and information flow across DHHS.

So taken together, the pattern suggests that the DHHS network contains both institutionally-driven coordination (status homophily) and voluntary, cross-agency collaboration (inbreeding homophily). Nodes with very high heterogeneity likely play "boundary-spanning" roles, actively integrating information across agencies: an interpretation consistent with Burt's (2004) argument that actors bridging structural holes are typically engaging in purposeful collaboration rather than mechanical or mandated coordination.

Overall, the pattern implies that the DHHS network is not simply segmented into agency-based clusters; instead, cross-agency ties are common, and a sizeable share of actors function as everyday brokers connecting different organizational units.

Task 2: Homophily Descriptive Measures - Newman Assortativity

Define (~200 words)

- a. Newman's Assortativity measures the tendency of nodes to create ties with others that are similar based on attributes (Lecture 5 Slides, p.34). The most common form is the assortativity coefficient (r), the Pearson correlation of attributes, ranging from -1 (full heterophily) to $+1$ (full homophily) and 0 represents random mixing. When calculated on node degree, it becomes degree assortativity, describing whether high-degree nodes connect to other high-degree ones (Newman, 2002, p.208701-2).
- b. Newman assortativity assumes that the observed mixing pattern can be meaningfully compared to a baseline of random mixing, meaning edges are formed independently of node attributes except through the assortative tendency being measured. Newman assortativity's main strength is that it provides a clear, single, and comparable coefficient that captures whether similar nodes tend to connect with each other, making it easy to detect systematic mixing patterns across very different networks (Newman, 2003). Newman shows that this measure reliably distinguishes strong empirical regularities, for example, social networks consistently exhibit positive assortativity, whereas technological and biological networks tend to be disassortative, demonstrating its usefulness for comparative analysis across domains (Newman, 2003, p.7-8). Moreover, assortativity has meaningful structural implications: assortatively mixed networks percolate more easily and are more robust to vertex removal, giving the measure explanatory power beyond simple description (Newman, 2003, p.10). A key limitation of Newman's assortativity measure is that observed mixing can arise from structural or finite-size artifacts rather than real preference: for example, disassortativity may appear simply because high-degree nodes cannot form multiple edges with one another (Newman, 2003, p.5-6). In addition, the coefficient reduces the entire mixing matrix into a single number, losing details about which specific categories or degrees mix more or less strongly (Newman, 2003). Finally, Newman notes that similar assortativity values can result from very different social, organizational, or structural mechanisms, meaning the

measure cannot explain why mixing occurs and should not be over-interpreted as evidence of a single underlying process (Newman, 2003, p.10-12).

- c. Unlike the Blau Index (ego-level diversity), the assortativity coefficient is a whole-network measure: one r value summarizes overall mixing.
- d. Assortativity: interpreted as regular homophily, calculated by attribute (e.g., probability of tie, given same attribute); Degree Assortativity: assesses whether nodes with high degree are more likely to connect with other high degree nodes; that's not homophily in the classical sense, which is by attribute (Lecture 5 Slides, p.40)! The difference between assortativity and degree assortativity matters because they measure fundamentally different mechanisms of tie formation. Attribute assortativity captures homophily in the sociological sense: whether people connect based on shared attribute such as gender, race, or organization. In contrast, degree assortativity is about topology, reflecting structural preference or hierarchy in the network (e.g., elites connecting with elites). Mixing by degree can therefore produce positive or negative correlations that have nothing to do with preference for similar people. If we confuse the two, we may misinterpret structural features of the network as evidence of social homophily, leading to incorrect conclusions about why ties form.
- e. I find Newman's assortativity most appropriate when analyzing overall social segregation or mixing—for example, gender homophily in friendship networks. I would use it when I need a single summary statistic of network-level homophily to compare across contexts or time. My overall assessment: it is a powerful, intuitive measure for detecting social mixing patterns, but it should be complemented by node-level diversity metrics (like Blau Index) for a more complete understanding of social structure.

Compute

```
# assortativity
g_dhhs <- asIgraph(DHHS)
vertex_attr_names(g_dhhs)

## [1] "agency"      "na"          "vertex.names"
table(V(g_dhhs)$agency)

##
##  0  1  2  3  4  5  6  7  8  9 10
##  2  4 12  2  2  3  2 16  3  5  3
agency_types <- as.integer(as.factor(V(g_dhhs)$agency))

assortativity_nominal(g_dhhs,
                      types    = agency_types,
                      directed = FALSE)

## [1] 0.1856224

# degree assortativity
assortativity_degree(g_dhhs)

## [1] -0.2123333
```

Interpret

Assortativity (Newman 2002) measures the tendency of nodes to connect with others that are similar to themselves in some attribute. When based on categorical attributes (like agency), it reflects status homophily: ties formed between actors sharing the same institutional status. When based on degree, it captures structural assortativity: whether highly connected nodes tend to connect with other highly connected nodes.

In my DHHS collaboration network, the nominal assortativity by agency is $r = 0.186$, a positive but modest value. This suggests a mild tendency for actors from the same agency to collaborate more frequently than random expectation, indicating the presence of status-based homophily, that is, communication and coordination remain somewhat structured along organizational boundaries. However, because the coefficient is far below 1, cross-agency collaboration is still relatively common, implying organizational boundaries matter but collaboration is also required across units. This aligns with Newman’s finding that assortativity in social systems is rarely extreme but often reflects the partial influence of organizational categories rather than strict segregation.

In contrast, the degree assortativity is $r = -0.212$, indicating disassortative mixing by degree. This means that highly connected “hub” individuals tend to link with less-connected ones rather than with other hubs. Substantively, the negative degree assortativity reflects a core–periphery collaboration pattern (Newman, 2003), suggesting that inter-agency collaboration depends on a few key bridging actors instead of evenly distributed, peer-to-peer connections.

Taken together, the findings describe a network consistent with Newman’s theoretical expectations: weak positive assortativity by agency indicates modest status-based clustering (homophily), while negative degree assortativity points to a structurally centralized, coordination-driven pattern in which high-degree actors play a crucial integrative role across agencies.

Task 3: ERGMs

Define (~200 words)

- ERGMs are models for the entire network: model the probability of a tie between two nodes (y), given other ties or attributes in the network (x) (Lecture 6 Slides, p.31). Instead of assuming ties are independent, ERGMs explicitly model dependence among edges, recognizing that relationships often form because of structural tendencies such as reciprocity, triadic closure, or homophily.
- The core idea is to represent the observed network as one possible realization among many, and estimate parameters that make such structures (e.g., mutual ties, triangles, attribute matches) more or less likely. Each parameter reflects how strongly a certain pattern contributes to the likelihood of tie formation. ERGMs are typically fitted using Markov Chain Monte Carlo (MCMC) methods because the number of possible networks grows exponentially with network size.
- Assessment: The major strength of ERGMs lies in their ability to capture the interdependence of social ties, moving beyond the dyad-independence assumptions of QAP or standard regressions. So ERGMs can capture structural patterns that depend on relationships between ties (Lecture 6 Slides, p.45) and test theoretical mechanisms within networks (e.g., “Do people prefer reciprocated friendships?”). However, ERGMs have difficulties in estimating bigger networks and multiple features (Lecture 6 Slides, p.51). This is mainly because the ERGMs are computationally intensive and highly sensitive to model specification.
- Overall, ERGMs are most appropriate when one wants to test hypotheses about the processes shaping network formation rather than just describe network patterns.

Compute

- Model 1: edges only (baseline)

```
set.seed(42)

mod1 <- ergm(middleschool ~ edges)
summary(mod1)

## Call:
## ergm(formula = middleschool ~ edges)
```

```
##
## Maximum Likelihood Results:
##
##      Estimate Std. Error MCMC % z value Pr(>|z|)
## edges -1.86274    0.08033      0 -23.19  <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 1847  on 1332  degrees of freedom
## Residual Deviance: 1051  on 1331  degrees of freedom
##
## AIC: 1053  BIC: 1059  (Smaller is better. MC Std. Err. = 0)
```

```
gden(middleschool)
```

```
## [1] 0.1343844
```

b. Model 2: edges + nodematch

```
set.seed(42)
```

```
mod2 <- ergm(middleschool ~ edges + nodematch("gender"))
summary(mod2)
```

```
## Call:
## ergm(formula = middleschool ~ edges + nodematch("gender"))
##
## Maximum Likelihood Results:
##
##      Estimate Std. Error MCMC % z value Pr(>|z|)
## edges      -5.3891    0.5785      0 -9.316  <1e-04 ***
## nodematch.gender  4.3530    0.5851      0  7.439  <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 1846.5  on 1332  degrees of freedom
## Residual Deviance:  811.2  on 1330  degrees of freedom
##
## AIC: 815.2  BIC: 825.6  (Smaller is better. MC Std. Err. = 0)
```

c. Model 3: edges + nodematch + gwesp(0.5, fixed = TRUE)

```
set.seed(42)
```

```
mod3 <- ergm(middleschool ~ edges + nodematch("gender") +
              gwesp(0.5, fixed = TRUE))
summary(mod3)
```

```
## Call:
## ergm(formula = middleschool ~ edges + nodematch("gender") + gwesp(0.5,
##      fixed = TRUE))
##
## Monte Carlo Maximum Likelihood Results:
##
##      Estimate Std. Error MCMC % z value Pr(>|z|)
## edges      -5.5348    0.5581      0 -9.918  <1e-04 ***
## nodematch.gender  2.7651    0.5769      0  4.793  <1e-04 ***
```

```
## gwesp.OTP.fixed.0.5  0.9288      0.1061      0  8.757  <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Null Deviance: 1846.5 on 1332 degrees of freedom
## Residual Deviance: 744.9 on 1329 degrees of freedom
##
## AIC: 750.9 BIC: 766.4 (Smaller is better. MC Std. Err. = 0.8294)
```

d. Model 4

```
set.seed(42)

mod4 <- ergm(middleschool ~ edges + nodematch("gender") +
             gwesp(0.5, fixed = TRUE) +
             mutual)

summary(mod4)

## Call:
## ergm(formula = middleschool ~ edges + nodematch("gender") + gwesp(0.5,
## fixed = TRUE) + mutual)
##
## Monte Carlo Maximum Likelihood Results:
##
##              Estimate Std. Error MCMC % z value Pr(>|z|)
## edges          -5.5052     0.5592      0 -9.846  <1e-04 ***
## nodematch.gender  2.5228     0.5886      0  4.286  <1e-04 ***
## gwesp.OTP.fixed.0.5  0.6833     0.1068      0  6.397  <1e-04 ***
## mutual           1.9337     0.3045      0  6.350  <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Null Deviance: 1846.5 on 1332 degrees of freedom
## Residual Deviance: 697.5 on 1328 degrees of freedom
##
## AIC: 705.5 BIC: 726.2 (Smaller is better. MC Std. Err. = 0.7014)
```

Model 4 builds on Model 3 by adding a mutual term, aiming to capture the strong reciprocity typically observed in adolescent friendship networks.

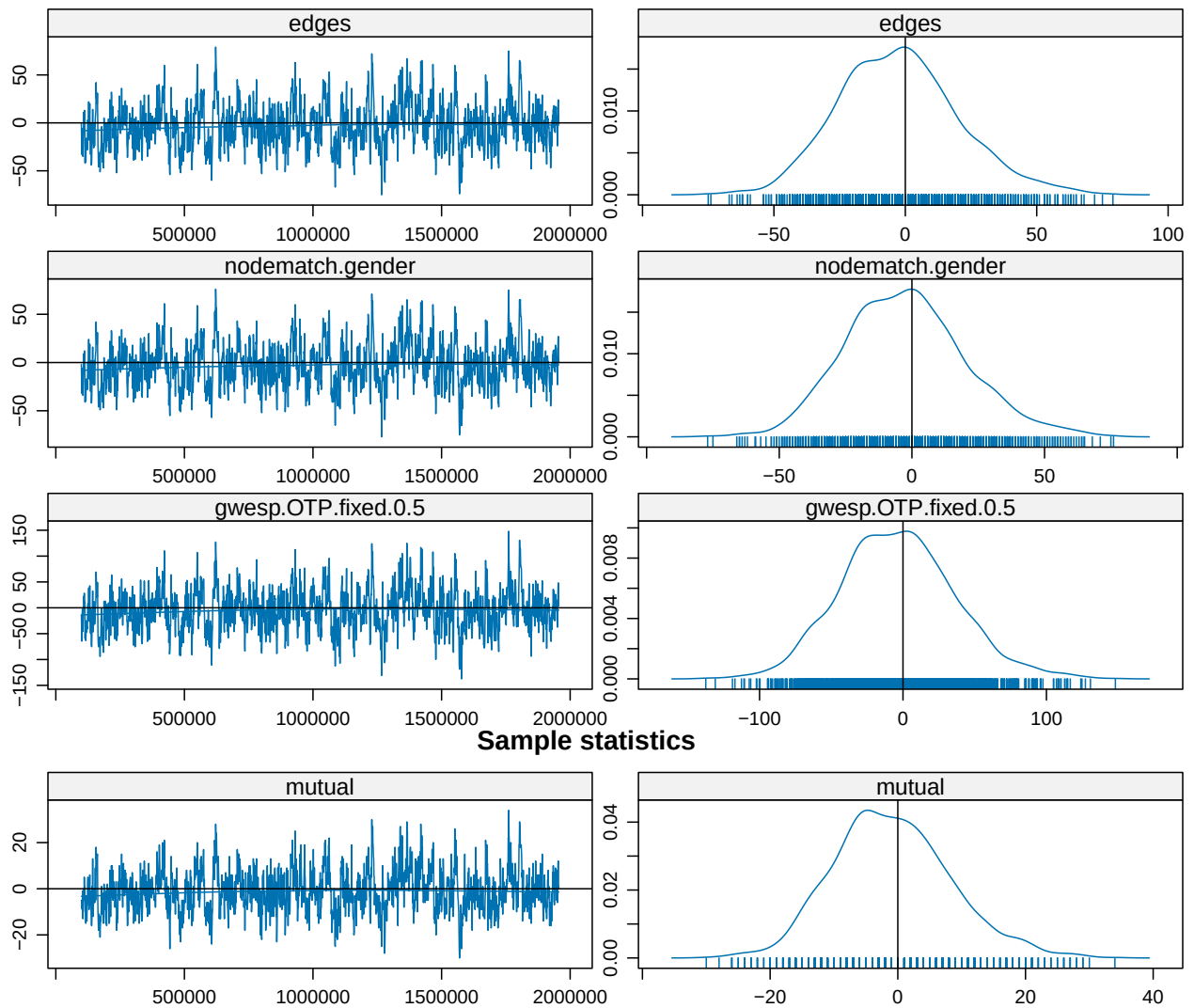
Hypothesis: Friendship ties in adolescent networks are often reciprocal. When a student nominates another as a friend, the likelihood of being nominated back is high due to shared daily interactions and social reinforcement. Therefore, I hypothesize that adding a mutual effect will significantly improve model fit.

Results: Compared to Model 3 (AIC = 750.9, BIC = 766.4), Model 4 (AIC = 705.5, BIC = 726.2) shows a substantially better fit, confirming that reciprocity plays a crucial role. All terms remain statistically significant ($p < 0.001$). The positive and significant coefficient for mutual (1.93) indicates that mutual nominations are far more likely than by chance, consistent with the expectation that friendship ties are often bidirectional in middle school settings.

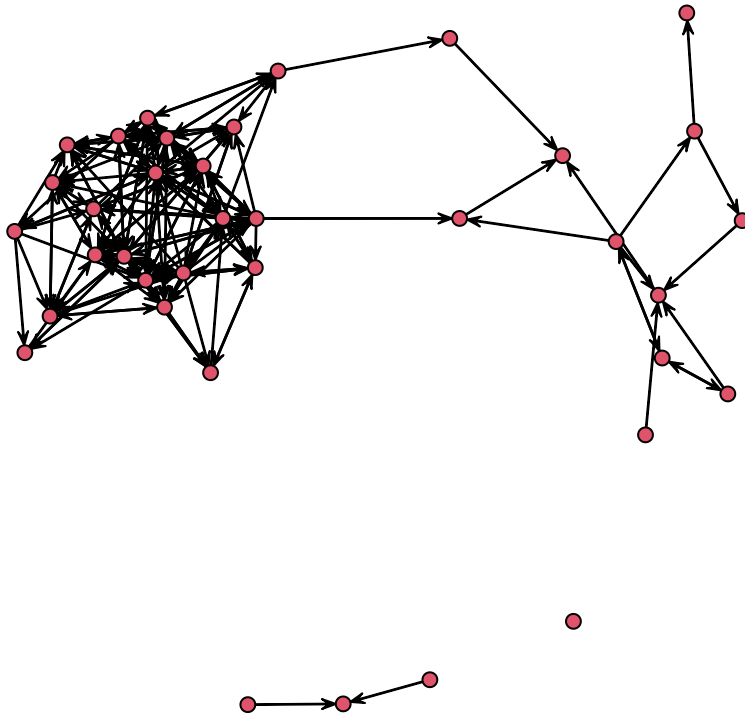
Thus, including the mutual effect improves both model convergence and substantive interpretability.

```
set.seed(42)
invisible(capture.output(mcmc.diagnostics(mod4)))
```

Sample statistics

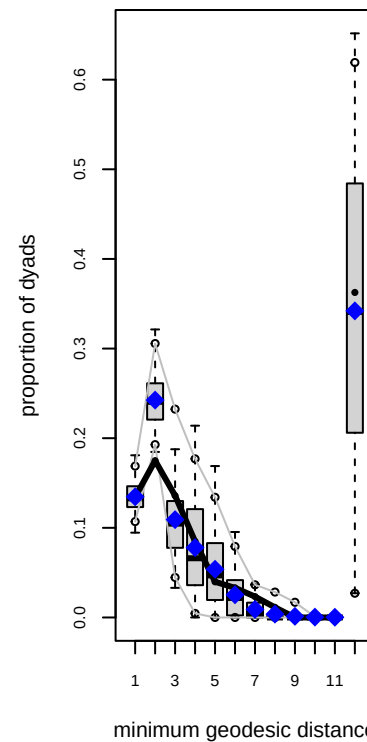
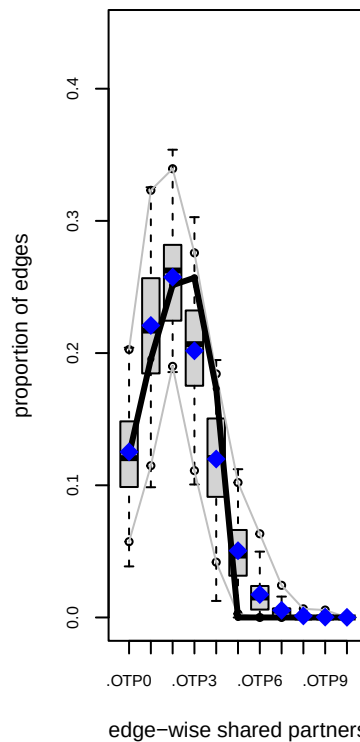
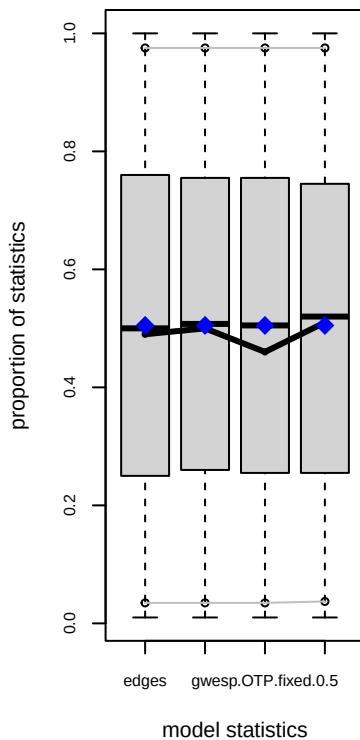


```
plot(mod4$newnetwork)
```

```
par(mfrow=c(1,3))
gof <- gof(mod4 ~ distance + esparkers + model)
plot(gof)
```

Goodness-of-fit diagnostics



The ERGM (mod4) for the Middle School friendship network successfully converged. MCMC diagnostics indicate stable parameter chains and a joint p-value of 0.66, suggesting no convergence issues. The GOF plots

show that observed network statistics (blue diamonds) fall within the simulated distributions (gray boxes), confirming that the model adequately reproduces key structural features of the network.

Interpret

a. model's coefficients

In Model 1, the only effect included is the edges term, which captures the baseline tendency for any two students to form a tie.

The negative coefficient (-1.86) indicates a low baseline probability of friendship between any two students, meaning the network is sparse. The density is 0.134 , that means, holding everything else constant, any random pair of students has only about a 13% chance of forming a tie. The p-value is < 0.0001 , showing that the edges term is statistically significant, reflecting a real and reliably estimated level of sparsity in the observed network.

This tells us that the middleschool network is relatively sparse, and students do not connect at random but that additional social processes must be shaping the structure, which aligns with typical adolescent networks, where relationships tend to cluster by classroom, friendship groups, or gender (here is gender).

The AIC (1053) and BIC (1059) reflect the model's goodness of fit. Because Model 1 only includes a single parameter, these values are relatively high, indicating that an edges-only ERGM provides a poor fit to the observed data. That means, an edges-only model cannot capture the mechanisms that actually generate ties, such as reciprocity, gender, popularity, or transitivity. The large difference between the null deviance (1847) and residual deviance (1051) also indicates that while the model reduces deviance, it is still missing key social processes, which later models will need to capture.

In Model 2, I add a nodematch (gender) term to test gender homophily in the middleschool network.

The edges coefficient becomes -5.39 , which is much more negative than Model 1's -1.86 . This change shows that once gender homophily is accounted for, the baseline tendency to form a tie (which now reflects cross-gender ties) becomes extremely low. The p-value (< 0.0001) means the model is statistically significant that this negative baseline is not due to chance, cross-gender ties truly have very low probability.

The nodematch (gender) coefficient is 4.35 , large and highly significant ($p < 0.0001$). This means that students are much more likely to form ties with peers of the same gender: holding all other structural processes constant, the log-odds of a tie increase by 4.35 when two students share the same gender. The strong significance indicates that this pattern is one of the key, systematic forces shaping the network rather than noise.

This tell us that gender homophily is a dominant mechanism in the middleschool network. Same-gender ties are strongly favored, while cross-gender ties are very unlikely. This finding aligns with established patterns in adolescent networks, where gender often structures friendship formation. For GOF, adding gender homophily dramatically improves model fit (AIC/BIC drop sharply), meaning gender explains a substantial amount of the network's structure, so ignoring gender (as in Model 1) hides a major part of the mechanism behind the network's structure.

One limitation is that Model 2 still includes no structural effects: Firstly, Model 2 cannot account for the high local clustering typically seen among adolescents: friends of friends become friends. Secondly, Model 2 does not consider reciprocity but actually friendships are rarely one-sided. It may distort estimates of other effects, such as exaggerating the gender coefficient because unmodeled reciprocity can be conflated with homophily. Finally, while the fit improves (lower AIC/BIC), the deviance remains relatively high, indicating that important sources of network structure remain unexplained. Therefore, Model 2 should be seen as capturing one important mechanism, but not a comprehensive model of tie formation in the middleschool network.

In Model 3, it shows that the middleschool network is shaped by both attribute-based processes (gender homophily) and structural processes (triadic closure).

The edges coefficient is -5.53 , slightly more negative than Model 2's -5.39 . This change indicates that after accounting for both gender homophily and triadic closure (gwesp), the baseline probability of a tie (now interpreted as the probability of a cross-gender, non-triadic tie) becomes even lower. The p-value (< 0.0001) confirms that this negative baseline is statistically meaningful and not due to random variation: cross-gender ties without structural support are really rare in this network.

The nodematch (gender) coefficient decreases from 4.35 (Model 2) to 2.77 (Model 3). This drop tells us that some of the apparent gender homophily observed in Model 2 was actually capturing unmodeled clustering due to shared friends. Once triadic closure is included in Model 3, part of what looked like “same-gender preference” is now correctly attributed to triangles and group formation. However, the coefficient remains positive, large, and highly significant ($p < 0.0001$), meaning gender homophily is still a strong mechanism, even after accounting for structural clustering. Gender still matters, but not as overwhelmingly as Model 2 suggested.

The gwesp (0.5) coefficient is 0.93 , positive and highly significant ($p < 0.0001$), capturing the tendency for “a friend of my friend is also my friend (holding other effects constant, the log-odds of a tie increase by 0.93).” This finding is highly consistent with adolescent friendship networks, where students frequently form small, cohesive friendship circles. The strong significance shows that triadic closure is a key structural mechanism, not random noise.

So taken together, this model recovers a more realistic picture of adolescent social structure than Model 2: friendship formation is driven by both who you are (gender) and who your friends already are (closure).

However, one limitation is that Model 3 does not include reciprocity. In a directed friendship network like this middleschool dataset, mutual ties are a core social mechanism: friendships are rarely one-sided. This omission may cause Model 3 to misattribute some mutuality-driven ties to other effects. So in the next model (Model 4), I add an additional term to illustrate this point.

In Model 4, I add a mutual term, allowing the model to capture reciprocity in directed friendship ties. This gives Model 4 the most realistic and complete representation of middle-school friendship dynamics among all four models.

The edges coefficient is -5.51 , similar to Model 3's -5.53 . This indicates that the baseline probability of a tie is extremely low when students are of different genders, do not share friends, and do not reciprocate. The small change from Model 3 shows that adding reciprocity does not substantially alter the baseline. The p-value (< 0.0001) confirms this is a statistically reliable estimate.

The nodematch (gender) coefficient is 2.52 , slightly lower than Model 3 (2.77). This reduction means that once reciprocity is taken into account, some same-gender ties that looked like “homophily” in Model 3 were actually better explained by mutual friendships. However, the coefficient remains large and significant ($p < 0.0001$), indicating that gender homophily remains a strong mechanism even after controlling for structural processes.

The gwesp (0.5) coefficient decreases from 0.93 (Model 3) to 0.68 in Model 4. This drop indicates that some of the transitivity observed in Model 3 was partly driven by reciprocated ties, not solely triadic closure. But even after adjusting for reciprocity, gwesp remains positive and highly significant ($p < 0.0001$), showing that triadic closure is still a major mechanism: students tend to form friendship groups where “friends of friends become friends.”

The mutual coefficient is 1.93 (conditional on A to B, the log-odds of B to A increase by 1.93), positive and highly significant ($p < 0.0001$). This indicates a strong tendency for reciprocity in the friendship network. When student A names student B as a friend, the likelihood that B names A back increases substantially. This is a core feature of directed friendship data and matches theoretical expectations for adolescent friendships, where mutual recognition is common.

Model 4 indicates that tie formation in the middleschool network is shaped by multiple interacting mechanisms rather than a single dominant process, including gender homophily, triadic closure (“friends of friends become friends”), and reciprocity (friendships tend to be returned rather than one-sided), which all match what we expect in directed friendship data. These mechanisms suggest that the network is structured by

both attribute-based mechanism (gender homophily) and structural mechanism (clustering and reciprocity). Overall, it provides the best-fitting and most realistic model.

b. model fit

```
library(texreg)
texreg::screenreg(
  list(mod1, mod2, mod3, mod4),
  custom.model.names = c("Model 1", "Model 2", "Model 3", "Model 4"))
```

```
##
## =====
##               Model 1      Model 2      Model 3      Model 4
## -----
## edges          -1.86 ***    -5.39 ***    -5.53 ***    -5.51 ***
##                (0.08)        (0.58)        (0.56)        (0.56)
## nodematch.gender          4.35 ***    2.77 ***    2.52 ***
##                (0.59)        (0.58)        (0.59)
## gwesp.OTP.fixed.0.5          0.93 ***    0.68 ***
##                (0.11)        (0.11)
## mutual                      1.93 ***
##                      (0.30)
## -----
## AIC             1053.31      815.20      750.86      705.46
## BIC             1058.51      825.59      766.45      726.24
## Log Likelihood  -525.66     -405.60     -372.43     -348.73
## =====
## *** p < 0.001; ** p < 0.01; * p < 0.05
```

From Model 1 to Model 2, AIC decreases from 1053 to 815.2 and BIC decreases from 1058.51 to 825.59, indicating a major improvement in model fit. This improvement happens because Model 2 introduces `nodematch` (gender), which captures gender homophily, a strong and well-known mechanism in adolescent friendship networks, whereas Model 1 only models overall density, which ignores the fact that same-gender ties are much more common than cross-gender ties in adolescent friendship. By adding gender homophily, Model 2 explains a large portion of the variation in tie formation that Model 1 could not account for. As a result, the likelihood improves significantly, leading to much lower AIC/BIC. Overall, the large drop in AIC/BIC shows that gender homophily is essential for explaining who forms ties with whom, and including it significantly increases the model's explanatory power.

From Model 2 to Model 3, AIC decreases from 815.20 to 750.86 and BIC decreases from 825.59 to 766.45, indicating another improvement in model fit. This improvement happens because Model 3 adds the `gwesp` (0.5) term, which captures triadic closure: the tendency for “friends of friends” to become friends. This is a core structural feature of adolescent friendship networks, where students form tightly knit groups and many ties occur inside local clusters. However, Model 2 could only explain who tends to connect (gender homophily), but not how friendships cluster. By modeling closure, Model 3 explains additional structure that Model 2 could not capture, leading to lower AIC/BIC. Overall, the improvement from Model 2 to Model 3 shows that friendships are shaped not only by gender but also by strong clustering tendencies, and incorporating structural dependence substantially enhances the model's explanatory power.

Comparing AIC and BIC across the three models shows a clear pattern of improvement: both AIC and BIC decrease. Because lower AIC/BIC indicates better fit, Model 3 fits the data best (because it includes both attribute-based mechanism: gender homophily and structural mechanism: triadic closure), followed by Model 2, and Model 1 fits the worst.

AI Disclosure Statement and Learning Reflections

I mainly used AI (ChatGPT) to help me debug. I originally worked with the DHHS network, but when I reached Model 4 of the ERGMs, the model kept running all the time but never returned results. I repeatedly asked AI for solutions and tried many different combinations, such as edges + nodematch + gwidegree, edges + nodematch + gwesp + gwidegree, and edges + nodematch + gwesp + gwdsp and so on. The only specification that successfully converged was edges + nodematch + nodefactor, but I felt that nodefactor did not provide meaningful substantive insights in this context. Because I think (but my thoughts could be wrong) in the DHHS network, we are more focus on identifying which nodes act as cross-agency bridges, not which agency is more “active.” Moreover, the goal of the ERGM is to explain tie-formation mechanisms, such as homophily, reciprocity, and triadic closure, rather than agency-level activity differences. For these reasons, I decided to switch to the middleschool network instead.

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